Chapter #6
Counting
Exercise 6.1

Q1 There are 18 mathematics major and 325 computer science major at college:

a) In how many ways can two representatives be picked so that one is mathematics major and other is computer science major.

Product rule applies here:-

$18 \times 325 = 5850$ ways.

b) In how many ways one representative be picked who is either mathematics major or computer science major?

Sum rule:

$18 + 325 = 343$ ways

Q2 An office building contains 27 floors and has 37 offices on each floor. How many offices are in buildings?

Product rule

$27 \times 37 = 900$ offices.

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3: A multiple-choice test contains 10 questions. There are four posible answers for each question.

a) In how many ways can a student answers the question on the test if student answers every question?

Product rule applies, since student will perform each 10 questions, one after other. There are 4 way to do each question. Therefore there

$4^{10} = 1,048,576$ ways to answer the questions.

b) In how many ways can a student answers the question on the test if student can leave answers blank?

This is identical to part @, except that now there are 5 ways to each answers so there are

$5^{10} = 9,765,625$ ways to answer the questions.

4: A particular brand of shirt comes in 12 colors, has a male versions and a female version, and comes in three sizes for each sex. How many different types of this shirt are made?

Product rule:

$12 \cdot 2 \cdot 3 = 72$

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5: Six different airlines fly from New York to Denver and seven fly from Denver to San Francisco. How many different pairs of airlines can you choose on which to book a trip from New York to San Francisco via Denver, when you pick an airline for flight to Denver and an airline for continuation flight to San Francisco?

Product rule:

$6 \cdot 7 = 42$ possibilities.

6: There are four major auto routes from Boston to Detroit and 6 from detroit to Los Angeles. How many major auto routes are there from Boston to Los Angeles via Detroit?

Product rule:

$4 \cdot 6 = 24$ routes.

7: How many different three-letter initials can people have?

Three letters are determined by specifying the first initial (26) then second (26) then third (26): so

Product rule: -

$26 \cdot 26 \cdot 26 = 17,576$ ways

8: H
ini

9: Ho
a

10: H
a

11: H
ba
th
r
th
fi

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8: How many different three letter initials with none of the letter repeated can people have?

There are 26 choices for first initial, 25 choices for second and 24 for thirds. Therefore

Product rule:

$26 \cdot 25 \cdot 24 = 15600$

9: How many different three-letters initials are there that begin with an A.

First initial is A. 1, 2nd - 26, 3rd - 26

Product rule:

$1 \cdot 26 \cdot 26 = 676$ ways

10: How many bit strings are there of length eight?

We have two choices for each bit, so there are $2^8 = 256$ bits.

11: How many bit strings of length 10 both begin and end with a 1.

A bit string is determined by choosing the bits in string, one after another, so product rule applies. We want to count the number of bit string of length 10, except that we are not free to choose either the first bit or last bit. Therefore, there are 8 choices to make with two ways so

$2^8 = 256$ strings.

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12. How many bit strings are there of length six or less, not counting the empty string?

Sum Rule, adding number of bit strings of each length up to 6. if we include emply string, then we ge

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 = 2^7 - 1 = 127$$

(Theorem 1 in section 2.4)

13: How many bit strings with length not exceeding n, where n is positive integer, consist entirely of 1s, not counting empty string?

This is a trick question, since it is easier than one might expect. Since string is given to consist entirely of 1's, there is nothing to choose except length. Since there are n+1 possible length not exceeding n, the answer is n+1.

14: How many bit strings of length n, where n is a positive integer, start and end with 1's?

if n=0, then the empty string vacuously satisfies condition.

If n=1, then there is one, namely the string 1.

If $n \geq 2$, then such string is determined by specifying the n-2 bits b/w the first bit and last bit, so there are

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2^{n-2} such strings.

15: How many strings are there of four lower case letters of length four or less, ~~not counting the empty string~~? that have letter x in them?

We can subtract from the number of strings length 4 of lower cas letters the number strings of length 4 of lower case letters other than x.

Thus $26^4 - 25^4 = 66,351$.

16: Hao many strings are there of four lower case letters of length 4 or less, not counting empty string?

By sum rule, we can count the number strings of length 4 or less by counting number of strings of length i, for $0 \leq i \leq 4$ and then adding results. Since there are ~~4+1~~ 26 letters ~~possible length not exceed~~ to choose from, and a string of length i is specified by choosing its characters, one after another. Therefore produte rule, 26^i strings of length i. answer is thus

$$\sum_{i=0}^{4} 26^i = 1 + 26 + 676 + 17576 + 45976$$
$$= 475,255.$$

20: How many positive integers b/w 5 and 31?

Because neither 5 nor 31 is divisible by either 3 or 4, whether ranges are

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meant to be inclusive or exclusive of their endpoints is moot.

a) are divisible by 3? which are these?

$\lfloor 31/3 \rfloor = 10$, $\lfloor 5/3 \rfloor = 1$. $10 - 1 = 9$.

6, 9, 12, 15, 18, 21, 24, 27, 30.

b) are divisible by 4? which are these?

$\lfloor 31/4 \rfloor = 7$, $\lfloor 5/4 \rfloor = 1$. $7 - 1 = 6$

8, 12, 16, 20, 24, 28

c) are divisible by 3 and 4, which are these?

both 3 and 4 = $4 \times 3 = 12$

$\lfloor 31/12 \rfloor - \lfloor 5/12 \rfloor = 2 - 0 = 2$.

12, 24

21: How many positive integers b/w 50 and 100?

Because neither 50 nor 100 is divisble by 7 or 11, whether ranges are meant to be inclusive or exclusive of their endpoints is moot.

a) are divisible by 7? which are these?

$\lfloor 100/7 \rfloor = 14$, $\lfloor 50/7 \rfloor = 7$, $14 - 7 = 7$

56, 63, 70, 77, 84, 91, 98.

b) are divisible by 11? which are these?

$\lfloor 100/11 \rfloor = 9$, $\lfloor 50/11 \rfloor = 4$. $9 - 4 = 5$

55, 66, 77, 88, 99.

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c) are divisible by 7 and 11, which are these?

both 7 and 11. $7 \times 11 = 77$

$$\lfloor 100/77 \rfloor - \lfloor 50/77 \rfloor = 1 - 0 = 1$$

77

22: How many positive integers less than 1000.

a) are divible by 7

$$\lfloor 999/7 \rfloor = 142$$

b) are divisible by 7 but not 11?

$\lfloor 999/11 \rfloor = 90$.

$7 \times 11 = 77$

$\lfloor 999/77 \rfloor = 12$

$= \lfloor 999/7 \rfloor - \lfloor 999/77 \rfloor$

$= 142 - 12$

$= 130$

c) divisible by 7 and 11

$7 \times 11 = 77$

$\lfloor 999/77 \rfloor = 12$

d) divisible by 7 or 11?

divicible 7 + divicible 11 - divisible by both

$= 142 + 90 - 12$

$= 220$

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e) exactly one of 7 and 11

divisble by 7 or 11 - divisible by both

a) - 220 - 12

= 208

f) divisible by neith 7 nor 11?

b) ~~divisble by~~

Number - divisble by 7 or 11

= 999 - 220

= 779

c) ~~go have distinct digit~~?

23) How many positive number b/w 100 and 999.

21 a) divisible by 7

$\lfloor 99 \rfloor 7 \rfloor = 14$, $\lfloor 999/7 \rfloor = 142$

142 - 14 = 128.

b) are odd.

$\lfloor 999/2 \rfloor = 499$

999 - 499 = 500

$99 - \lfloor 99/2 \rfloor = 50$

500 - 50 = 450

not divisble by 4

d) ~~have same three desired digits~~?

1 $999 - \lfloor 999/4 \rfloor = 750$

$99 - \lfloor 99/4 \rfloor = 75$

750 - 75 = 675

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c) have same three-decimal digits?

There are just 9 digits that a three-decimal can start with. If all the digits are to be same, then there is no choice after the leading digit has been specified. Therefore there are 9 such numbers.

e) are divisibe by 3 or 4.

$\lfloor 999/3 \rfloor - \lfloor 99/3 \rfloor = 300$

$\lfloor 999/4 \rfloor - \lfloor 99/4 \rfloor = 225$

$\lfloor 999/12 \rfloor - \lfloor 99/12 \rfloor = 75$

$300 + 225 - 75 = 450$

f) not by 3 or 4

$900 - 450 = 450.$

g) divisible by 3 but not 4.

divisble by 3 - divisble by both 3 and 4

$\therefore 300 - 75$

225

h) divisble by 3 and 4.

75 digits.

24. How many positive integer b/w 1000 and 9999

a) are divisible by 9?

Every ninth number is divisble by 9, so answer is one ninth of 9000 or 1000.

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b) are even?

Every other number is even, so answer is one half of 9000 or 1000.

c) are not divisble by 3.

Every third number, so one third of 9000 or 3000 numbers in this range are divisble by. The remaining 6000 are not.

d) are divisible by 5 or 7?

1800 divisible by 5

1286 divisible by 7

257 by 5 and 7

so

$1800 + 1286 - 257 = 2829$

e) not divisible by 5 or 7?

$9000 - 2829 = 6171$

f) are divisible by 5 not by 7

$1800 - 257 = 1543$

g) are divisible by 5 and 7

257

25: How many strings of three-decimal digits

This problem involve 1000 posible strings since thire is choice of 10 digits for each of three position in string.

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a) do not contain same digit 3 times?

(000,111,222,333,444,555,666,777,888,999)

$1000 - 10 = 990$.

b) begin with odd

$5 \cdot 10 \cdot 10 = 500$.

c) have exactly two digits the are 4.

$3 \cdot 9 = 27$

27: A commitee is formed consisting of one representative from each of 50 states in US, where representative from the state is either governor or one of two senators from state. How many ways are there to form this commitee?

Product rule.

$3^{50} \approx 7.2 \times 10^{23}$

28: How many license platescan be made using either three digits by uppercase or lower uppercase letters followed by three digits?

$10^3 26^3 + 26^3 10^3 = 35152000$

29: How many license plate can be made using either two uppercase letters followed by 4 digits or two digits followed by 4 uppercase letters?

Product rule:-

$26^2 10^4 + 10^2 26^4 = 52,457,600$

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Exercise 6.3

2: List all the permutations of {a,b,c}

abc, acb, bca, bac, cab, cba

2: How many different permutations are there of the set {a,b,c,d,e,f,g}

$= 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$= 5040$

3: How many permutations of {a,b,c,d,e,f,g} end with a?

The 6th letter b through to g can be listed in all possible order

$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$= 720$

4: Let $S = \{1,2,3,4,5\}$

(a) List all the 3-permutations of S

(b) List all the 3-combination of S

(a) $n = 5 \quad r = 3$

$${}^nP_r = \frac{n!}{(n-r)!} = \frac{5!}{(5-3)!} = \frac{5!}{2!} =$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1}$$

$$= \frac{120}{2} = 60$$

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(b) $n = 5 \quad r = 3$

$$nC_r = \frac{n!}{r!(n-r)!} = \frac{5!}{3!(5-3)!}$$

$$\frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1}$$

$$= \frac{120}{12} = 10$$

We list them the following way
Each combination is listed without punctuation, in increasing order

123(132, 231, 213, 312, 321) 124(142, 241, 214, 412, 421) 125(152, 251, 215, 521, 512)
134(143, 341, 314, 413, 431) 135(153, 531, 513, 315, 351)
145(154, 451, 415, 514, 541)
234(243, 432, 423, 324, 342) 235(253, 352, 325, 532, 523)
245(254, 452, 425, 524, 542) 345(354, 435, 453, 534, 543)

Q5 Find the value of each of these quantities

a P(6,3)

$^6P_3 = 120$

(b) P(6,5)

$^6P_5 = 720$

(c) P(8,1)

$^8P_1 = 8$

(d) P(8,5)

$^8P_5 = 6720$

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(e) $P(8,8)$

$^8P_8 = 40320$

(f) $P(10,9)$

$^{10}P_9 = 3628800$

6. Find the value of each of these quantities

(a) $C(5,1)$

$^5C_1 = 5$

(b) $C(5,3)$

$^5C_3 = 10$

(c) $C(8,4)$

$^8C_4 = 70$

(d) $C(8,8)$

$^8C_8 = 1$

(e) $C(8,0)$

$^8C_0 = 1$

(f) $C(12,6)$

$^{12}C_6 = 924$

7. Find the number of 5-permutation of a set with nine elements

$P(9,5) = {}^9P_5 = 15,120$

8. In how many different ~~orders~~ orders can five runners finish a race if no ties are allowed

$P(5,5) = {}^5P_5 = 120$

9. How many possibilities are there for the win, place and show (first 2nd and 3rd) positions in a

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horse race with 12 horses if all the orders of finish are possible

$P(12,3) = {}^{12}P_3 = 1320$

10 There are six different candidates for governor of a state. In how many different orders can the names of the candidates be printed on ballon

$P(6,6) = 720$

11 How many bit strings of length 10 contain

(a) at most four 1s ?

$C(10,4) + C(10,3) + C(10,2) + C(10,1) + C(10,0)$

$= 210 + 120 + 45 + 10 + 1$

$= 386$

(b) at least four 1s ?

$C(10,4) + C(10,5) + C(10,6) + C(10,7) + C(10,8)$

$+ C(10,9) + C(10,10)$

$= 210 + 252 + 210 + 45 + 10 + 1$

$= 848$

(c) at least four 1s

(c) exactly four 1s

$C(10,4) = 210$

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(d) an equal number of 0s and 1s?

To have an equal number of 0's and 1's in this case means to have five 1's Therefore answer is $C(10,5) = 252$

12. How many bit strings of length 12 contain

(a) exactly three 1s?

$C(12,3) = {}^{12}C_3 = 220$

(b) at most three 1s?

$C(12,3) + C(12,2) + C(12,1) + C(12,0)$

$= {}^{12}C_3 + {}^{12}C_2 + {}^{12}C_1 + {}^{12}C_0$

$= 220 + 66 + 12 + 1$

$= 299$

(c) at least three 1s?

$C(12,3) + C(12,4) + C(12,5) + C(12,6)$
$+ C(12,7) + C(12,8) + C(12,9)$
$+ C(12,10) + C(12,11) + C(12,12)$

$220 + 495 + 792 + 924 + 792 + 495$
$+ 220 + 66 + 12 + 1$
$= 4017$

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an equal number of 0's and 1's

$C(12,6) = 924$

2) How many permutations of the letter ABCDEFG contain

(a) the string BCD?

A, B, E, F, ~~B,C,D~~ BCD

$P(5,5) = {}^5P_5 = 120$

(b) the string CFGA

~~B~~, B, D, E, CFGA

$P(4,4) = {}^4P_4 = 24$

(c) the string BA and GF

BA, GF, C, D, E

$P(5,5) = 120$

(d) the string ABC and DE

ABC, DE, F, G

$P(4,4) = 24$

(e) the strings ABC and CDE

ABC, CDE = ABCDE

ABCDE, F, G = $P(3,3) = 6$

(f) the string CBA and BED

~~GBAED, F, G~~

There is no permutation with both of these substrings, since B cannot be followed by both A and E at the same time

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